

From Hand-Crafted Rules to Learned Policies: Robust Continuous Finger-Motion Guidance of a Collaborative Robot through a Capacitive Tactile Skin

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Abstract. Capacitive tactile skins let an operator guide a collaborative robot by sliding a finger over the robot body, but the controllers that translate touch into motion are almost always hand-crafted: a pipeline of centroid tracking, thresholds, and mode heuristics whose dozens of tuned parameters are brittle against sensor drift, damaged cells, and gain variation. We present a learned alternative and, equally importantly, a way to *prove* that it is more robust. First, we record the rule-based controller as a teacher and train a lightweight spatiotemporal policy—a location-preserving convolutional encoder with per-channel soft-argmax keypoints, followed by a recurrent temporal head—that maps raw 10×10 capacitance sequences directly to 6-DOF end-effector velocity commands and interaction modes. A geometry-aware augmentation scheme (touch-pattern flips and translations with consistently transformed velocity targets) multiplies the effective coverage of a five-minute demonstration corpus. Second, we introduce a *faithful-replay robustness benchmark*: recorded episodes are re-played through the unmodified deployed implementations of both the rule pipeline and the learned policy while calibrated perturbations—baseline drift, dead taxels, gain error, and Gaussian noise—are injected into the capacitance stream, and both are scored against the clean-input teacher command. The learned policy matches the teacher on clean input (1.45 cm/s velocity RMSE, 92.6% mode accuracy) and degrades far more gracefully than the rules under drift, dead taxels (up to 20% of the array), and gain error, while a targeted noise experiment exposes an honest failure mode of temporal-difference input channels. The policy runs in 1.9 ms per inference on a desktop CPU with 0.23 M parameters, well within a 60 Hz control budget.

Keywords: Tactile sensing · Imitation learning · Human–robot interaction · Robustness evaluation · Capacitive skin

1 Introduction

A tactile skin wrapped around a robot link invites the most direct form of robot guidance imaginable: place a finger on the robot and slide it in the direction the tool should move. Systems that realize this interaction today do so with hand-crafted signal-processing pipelines. A touch centroid is extracted from the thresholded capacitance image, its frame-to-frame displacement is scaled into a velocity command, and a stack of auxiliary heuristics—pressure thresholds for push and pull modes, hold counters, hysteresis bands, grace periods—arbitrates between interaction modes. Such controllers work, and they work well enough to serve as a product feature. But they inherit the classic weaknesses of rule-based perception. Every threshold is tuned to one sensor, one mounting, one operator, and one day’s baseline; capacitive arrays drift with temperature and humidity, individual taxels die, and gain varies across fabrication batches. Each new robustness patch adds parameters that interact with the existing ones.

This paper asks whether a *learned* policy can replace the rule pipeline for continuous tactile guidance, and—the question that is usually left unanswered—how to demonstrate that the replacement is more than a costly imitation. Our answer has two parts.

An imitation pipeline that inherits the rules’ competence. We treat the deployed rule-based controller as a teacher. While an operator interacts with the skin, we record synchronized capacitance frames together with the teacher’s velocity command and mode decision, yielding labeled episodes without any manual annotation. A compact CNN-GRU policy is trained offline on these episodes. Its convolutional encoder deliberately avoids global average pooling: a per-channel soft-argmax (spatial softmax) layer preserves *where* each learned feature fires on the taxel grid, which is precisely the information that determines the commanded direction. A geometry-aware augmentation scheme exploits the physical symmetries of the sensor: touch patterns are flipped and translated on the grid while the velocity targets and hand-crafted auxiliary features are transformed consistently, multiplying a five-minute demonstration corpus by an order of magnitude.

A robustness benchmark that both pipelines take identically. Offline agreement with the teacher cannot show superiority over the teacher—by construction, perfect agreement only ties the rules. We therefore built a *faithful-replay benchmark*: recorded episodes are streamed through the *unmodified deployed implementations* of both controllers (including all stateful counters, smoothing filters, and confidence gates) while calibrated sensor perturbations are injected into the capacitance stream. Both controllers are scored against the clean-input teacher command at every frame. The benchmark validates its own fidelity: with zero perturbation, the replayed rule pipeline reproduces the recorded teacher command exactly on 89.8–97.3% of frames, with residual deviations confined to episode-boundary counter states.

On this benchmark the learned policy is not merely equal to its teacher. Under baseline drift it holds a 30–40% lower velocity error than the rules until both collapse at extreme drift; with 20% of taxels dead its error rises by only

32% while remaining $1.7\times$ below the rules; and it is more tolerant to gain error on both sides of unity. A targeted noise sweep also exposes a genuine weakness—temporal-difference input channels amplify uncorrelated per-frame noise—which we analyze and discuss rather than hide.

Contributions:

1. An end-to-end imitation pipeline that converts a deployed rule-based tactile guidance controller into training supervision for a continuous velocity policy, requiring no manual labeling.
2. A location-preserving policy architecture (spatial-softmax convolutional encoder + recurrent temporal head, 0.23 M parameters, 1.9 ms CPU inference) and a geometry-aware augmentation scheme that transforms tactile frames, auxiliary features, and velocity targets consistently.
3. A faithful-replay robustness benchmark for tactile interfaces, with four calibrated perturbation families, that scores the deployed implementations of rule-based and learned controllers under identical corruptions—and results showing the learned policy degrades more gracefully on three of the four families.

2 Related Work

Tactile skins for physical human–robot interaction. Large-area robot skins have progressed from rigid pressure pads to conformal capacitive, resistive, and multimodal arrays [5, 3, 15], including textile and fabric-based sensors that wrap curved robot links [18, 14, 19]. Most systems use the skin for safety (collision detection, contact localization) or for perception during manipulation [21, 13]. A smaller body of work uses skin contact as an *input channel*: touch-gesture classification on robot surfaces [12, 7] and continuous lead-through control via force/torque interpretation [9]. Continuous *velocity* guidance from a taxel array is typically realized with hand-crafted centroid-tracking controllers; we are not aware of prior work that both learns this mapping end-to-end from the raw taxel stream and quantifies its robustness against the rule-based controller it replaces.

Learning from demonstration and behavior cloning. Casting controller construction as supervised imitation is a long-standing idea [16, 1]. Our setting is unusual in that the demonstrator is itself a *program*, which sidesteps the multimodality that plagues human demonstrations [17] and makes plain regression well-posed, but bounds the achievable policy by the teacher’s competence on clean input. The value of learning then lies in generalization under input corruption—exactly what our benchmark measures. Related programmatic-teacher setups appear in autonomous driving [4] and locomotion [10], where privileged or scripted experts supervise sensor-space students. Spatial-softmax feature points originate in end-to-end visuomotor policy learning [11, 6]; we show they matter even on a 10×10 tactile grid, where global pooling destroys the location information that determines command direction.

Robustness evaluation of tactile pipelines. Sensor-corruption benchmarks are standard in vision [8] but rare in tactile control, where evaluations usually report accuracy on clean held-out data. Perturbation studies exist for tactile object recognition [20], yet controller-level comparisons under identical corruption are missing. Our replay harness is closest in spirit to log-replay evaluation in autonomous driving [2], transplanted to the tactile interface setting with perturbation models specific to capacitive arrays (baseline drift, dead taxels, gain error, thermal noise).

3 System Overview

3.1 Sensing Substrate and Control Loop

Our testbed is a capacitive tactile skin providing a 10×10 grid of taxels mounted on the end link of a 6-DOF collaborative arm (TM5-900). The skin outputs, per frame, a matrix of relative capacitance changes (percentage deviation from a per-session calibration baseline); touching the surface produces negative deviations roughly proportional to contact pressure. A three-frame moving average is computed on-device alongside the instantaneous signal. Frames are consumed by the controller at approximately 60 Hz. [PLACEHOLDER: One to two sentences on skin construction and readout electronics, with a citation to your hardware publication if available. Kept intentionally brief here: the hardware is not a contribution of this paper.]

3.2 The Rule-Based Teacher

The deployed controller—our teacher and our baseline—follows the canonical centroid-tracking design. Taxels below a touch threshold $\tau = -3.0$ are clustered by connected components; a pressure-weighted centroid (r_t, c_t) is tracked with hysteresis. The normalized centroid displacement $(\Delta r_t, \Delta c_t)$, after a deadband and gain, is mapped to end-effector velocity: column motion commands v_x , row motion commands v_z . Two pressure-gated modes overlay this mapping: a hard press (peak below a push threshold, displacement within a hold deadband for k consecutive frames) commands a *push* along $+v_y$, and a two-finger pinch commands a *pull* along $-v_y$, both with pressure-modulated speed. Release logic, grace periods, and axis locks complete the pipeline. In total the controller exposes roughly thirty tuned parameters. This is a competent, production-quality design—and every one of its thresholds silently assumes a clean, calibrated, undamaged sensor.

3.3 Teacher-Supervised Data Collection

During recording sessions an operator interacts freely with the skin while the teacher runs. At each control tick we store the instantaneous and averaged capacitance matrices, the teacher’s 6-DOF velocity command $\mathbf{v}_t^* \in \mathbb{R}^6$, its mode decision $m_t^* \in \{\textit{stop}, \textit{move}, \textit{push}, \textit{pull}\}$, and the hand-crafted intermediate features

[**PLACEHOLDER:** Architecture diagram: 4-channel 10×10 frame window $\rightarrow$ conv stem (3 layers, no pooling) $\rightarrow$ per-channel spatial softmax keypoints + mean intensities $\rightarrow$ fusion with auxiliary-feature MLP $\rightarrow$ GRU (128) $\rightarrow$ velocity head (6-DOF) + mode head (4-way). Redraw in the style of your group’s figures.]

Fig. 1. Policy architecture. A location-preserving convolutional encoder turns each tactile frame into soft-argmax keypoints and intensities; a recurrent head integrates the 16-frame window and emits a 6-DOF velocity command and an interaction mode.

(centroid, displacement, speed, peak and mean pressure, touch presence). The corpus used in this paper comprises **8 episodes, 18,551 frames (≈ 5.2 minutes)** across two sessions: free omnidirectional swiping, and a push/pull-focused session. No frame was manually labeled. Rotational axes were not exercised and are locked to zero throughout.

4 Learned Policy

4.1 Input Representation

The policy consumes a sliding window of $T=16$ frames. Each frame is a 4-channel 10×10 image: the instantaneous capacitance deviation, its 3-frame moving average, the temporal difference between consecutive instantaneous frames, and a binary touch mask obtained by thresholding the averaged channel at τ . Alongside the frames, an 11-dimensional auxiliary vector carries the teacher pipeline’s own intermediate features (normalized centroid position, raw and normalized displacement, speed, peak and mean active pressure, touch presence, and control-frame index). Feeding the teacher’s features to the student is deliberate: it gives the policy a warm start on the quantities the rules found useful, while the raw frames let it learn what the rules discard.

4.2 Location-Preserving Spatiotemporal Architecture

Frame encoder. Three 3×3 convolutions (32, 64, 64 channels, batch normalization, GELU) process each frame at full 10×10 resolution—no spatial pooling at any depth. A per-channel *spatial softmax* then reduces each of the 64 feature maps $F_k \in \mathbb{R}^{10 \times 10}$ to its expected image coordinate,

$$\mathbf{p}_k = \sum_{(i,j)} \text{softmax}(F_k/\theta)_{ij} (u_i, u_j), \quad u \in [-1, 1], \quad (1)$$

with a learned temperature θ . Concatenating the 64 keypoints with the 64 per-channel mean intensities yields a 192-dimensional descriptor that preserves *where* the contact is and *how hard* it presses. The alternative—global average pooling, common in compact tactile encoders—collapses location entirely; on a guidance task where direction is decoded from position change, this is the single most damaging design choice available, and we quantify its cost in Sect. 6.2.

Temporal head and outputs. Frame descriptors are fused with an MLP-encoded auxiliary vector and passed through a single-layer GRU (128 units). Two heads read the final hidden state: a velocity head regressing the normalized 6-DOF command, and a 4-way mode classifier. The full model has **234,955 parameters** and runs in **1.9 ms** per window on a desktop CPU—an order of magnitude inside the 16 ms control budget, leaving headroom for embedded deployment.

4.3 Geometry-Aware Augmentation

Five minutes of demonstration cannot cover the product space of contact locations, directions, and pressures. The sensor’s geometry, however, provides exact symmetries. During training each sampled window is, with probability 0.5 per axis, *flipped* on the taxel grid—and the transformation is propagated consistently through every affected quantity: a column flip negates v_x (and the r_z target, if present), mirrors the centroid column, and negates the column displacement features; a row flip acts analogously on v_z , r_x , and the row features. Windows are additionally *translated* by up to ± 2 taxels, bounded such that no active cell leaves the grid (targets unchanged, centroid features shifted), and perturbed with Gaussian noise scaled to a fraction of each channel’s training standard deviation. The flip group alone multiplies directional coverage by four; translation decouples the learned mapping from absolute contact position. Mode labels are invariant under all three transforms. Augmentation is applied only to training windows, never to validation data.

4.4 Training Protocol

Windows are split by *episode* (never within an episode): 6 training episodes yield 13,847 windows, 2 held-out episodes (one per session) yield 4,584 validation windows. Normalization statistics use training data only. The loss is a weighted sum of a Smooth- L_1 velocity term and a class-balanced cross-entropy mode term ($\lambda_{\text{mode}} = 0.2$), with per-sample weights that de-emphasize *stop* frames and emphasize the rarer *pull* class. We train with AdamW (learning rate 10^{-3} , weight decay 10^{-4} , batch 128), plateau-based decay, and early stopping; the reported model converged at epoch 6. Rotational output axes are constrained to zero, matching the recorded task space.

5 Faithful-Replay Robustness Benchmark

5.1 Design

The benchmark answers one question: *when the sensor misbehaves, which controller keeps translating operator intent correctly?* Three design decisions distinguish it from standard offline evaluation.

(1) *Deployed implementations, not re-implementations.* Both controllers are exercised through their unmodified runtime classes. Recorded episodes are streamed frame-by-frame into a sensor emulation layer; the rule pipeline runs with all its stateful machinery (hold counters, hysteresis, grace periods, smoothing), and the learned policy runs with its deployed buffering, normalization, and confidence gating. Nothing is idealized.

(2) *Ground truth from clean input.* The reference at frame t is the teacher command recorded on the *clean* sensor stream. Both controllers then receive an identically *perturbed* stream. A controller scores well if corruption does not change what it believes the operator wants—the operator’s finger, after all, is doing the same thing regardless of sensor health.

(3) *Self-validation.* At zero perturbation the replayed rule pipeline must reproduce its own recorded output. It does: 89.8% and 97.3% of frames match exactly on the two held-out episodes, with residuals traceable to stateful-counter initial conditions at episode boundaries. This closes the loop on harness fidelity before any perturbation is trusted.

5.2 Perturbation Families

Four families model the dominant failure modes of capacitive arrays. All are injected consistently into both the instantaneous and the moving-average channel (noise is correctly attenuated in the average, as the physical sensor stack would). **Baseline drift:** a uniform offset toward the touch threshold, modeling thermal/humidity drift between calibrations (0–3.0 units; the touch threshold is -3.0). **Dead taxels:** a random fraction of cells clamped to zero (0–20%), modeling broken electrodes. **Gain error:** multiplicative scaling (0.5–1.3 $\times$), modeling batch variation and mounting differences. **Gaussian noise:** i.i.d. per-cell noise ($\sigma \in [0, 4]$ units), modeling electromagnetic interference. Velocity error is reported as linear-axis RMSE against the clean-input teacher command; mode agreement as the fraction of frames whose canonical mode (*stop/move/push/pull*) matches.

6 Results

6.1 Offline Imitation Fidelity

On the two held-out episodes the policy attains a linear-velocity RMSE of **1.45 cm/s** against teacher commands whose magnitudes reach 10 cm/s (per-axis R^2 : 0.82 / 0.83 / 0.81 for $v_x/v_y/v_z$), and **92.6%** mode accuracy. Mode

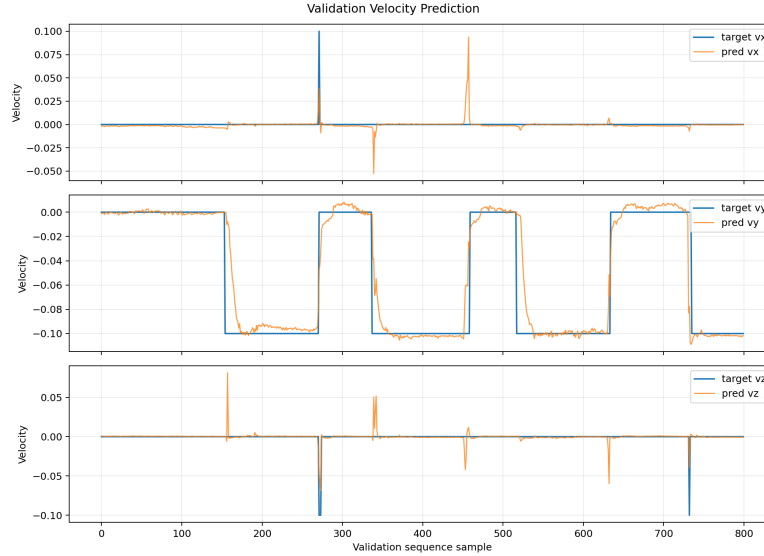


Fig. 2. Predicted vs. teacher velocity on held-out validation episodes (first 800 windows; v_x , v_y , v_z from top to bottom). The policy reproduces direction changes, push/pull events (v_y), and stop phases.

confusions concentrate where the teacher itself is ambiguous—*stop* frames bordering *push* onsets (Fig. 2). Note that these numbers measure agreement with the teacher; they establish that imitation succeeded, not that the policy is better. That question is answered next.

6.2 Robustness under Sensor Perturbation

Table 1 and Fig. 3 report the benchmark. Three findings stand out.

The learned policy wins three of four families. Under baseline drift up to 2.0 units—two-thirds of the way to the touch threshold—the policy’s error stays 27–38% below the rules’, and its mode agreement stays above 0.84 while the rules never exceed 0.71. With one in five taxels dead, the policy’s error rises only from 0.014 to 0.019 m/s; the convolutional encoder interpolates around missing cells, whereas the rules’ pressure-weighted centroid has no such mechanism. Under gain error the same picture holds on both sides of unity.

Both collapse at extreme drift, for the same reason. At drift 3.0 the offset equals the touch threshold: every idle cell registers as touched, touch segmentation fails, and neither controller receives a meaningful contact signal. This is a sensing failure, not a control failure, and marks the boundary beyond which recalibration—not robustness—is the remedy.

Uncorrelated noise is an honest weakness of the learned policy. Already at $\sigma=0.5$ the policy falls behind the rules. The cause is architectural: one input

Table 1. Faithful-replay benchmark on held-out episodes (4,614 frames): velocity RMSE (m/s, lower is better) and canonical-mode agreement (higher is better) against the clean-input teacher command. Both controllers receive identical perturbed streams. Bold marks the better controller per condition.

Perturbation	Level	RMSE (m/s)		Mode agree.	
		Rules	Learned	Rules	Learned
None (clean)	–	0.032	0.014	0.68	0.93
Baseline drift	1.0	0.032	0.020	0.69	0.90
	2.0	0.033	0.024	0.71	0.84
	3.0	0.052	0.062	0.48	0.07
Dead taxels	10%	0.032	0.015	0.68	0.93
	20%	0.033	0.019	0.67	0.88
Gain error	0.7×	0.036	0.019	0.61	0.90
	1.3×	0.029	0.016	0.75	0.92
Gaussian noise	$\sigma=0.5$	0.034	0.043	0.67	0.62
	$\sigma=2.0$	0.041	0.056	0.56	0.37

channel is a *temporal difference* of consecutive frames, which doubles the variance of i.i.d. noise, and the binary touch-mask channel flickers when cell values straddle the threshold. The rules are partially shielded because they read only the 3-frame moving average. The failure mode is thus attributable to input design rather than to learning per se—and it is addressable: training with realistic (temporally correlated) noise models, or low-pass filtering the instantaneous channel at deployment, are direct remedies that we discuss in Sect. 7.

A note on the clean-input mode gap. The rules’ modest clean-input mode agreement (0.68) is not an error of the replay: on a subset of recorded frames the ground-truth command reflects manually specified teaching intent that overrides the raw rule output, and the learned policy—trained on the intended command—captures this intent where the raw rules diverge from it. The learned policy thus imitates what the operator *meant*, not merely what the pipeline *did*.

6.3 Ablation: Location-Preserving Encoding and Augmentation

Table 2 compares the proposed configuration against a variant with a global-average-pooling encoder and no geometric augmentation. Clean-input fidelity is identical—on in-distribution data, either encoder suffices—but under perturbation the difference is large: at $\sigma=0.25$ noise the proposed configuration’s error is 40% lower. Robustness, not clean accuracy, is where the architectural and augmentation choices pay off; evaluations that report only clean held-out metrics would miss this entirely.

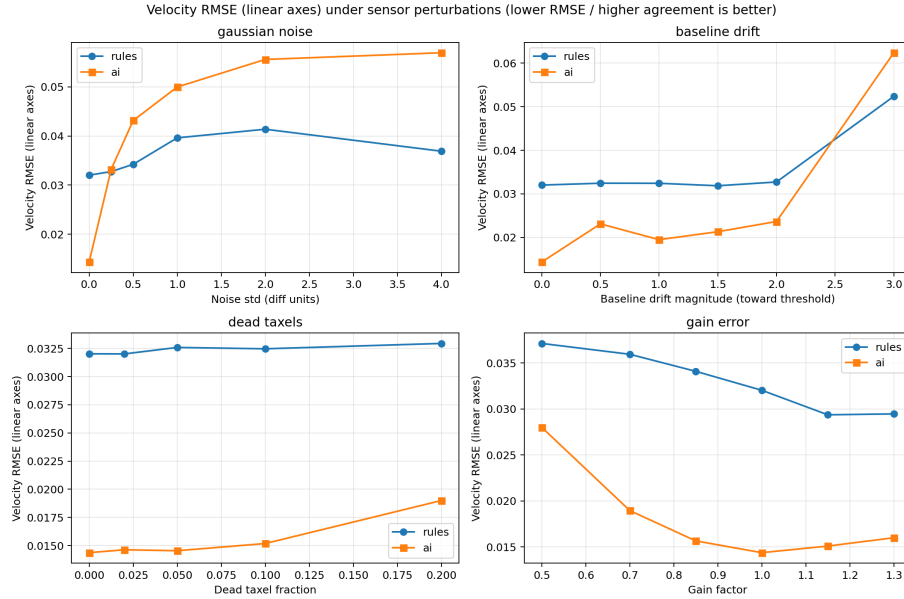


Fig. 3. Velocity RMSE against the clean-input teacher command as a function of perturbation magnitude, for the rule pipeline (circles) and the learned policy (squares), across all four perturbation families.

7 Limitations and Future Work

The teacher bounds the policy on clean input. Behavior cloning from a programmatic teacher cannot exceed the teacher where the teacher is right; our claim is superiority under corruption, not on clean data. Surpassing the teacher outright requires richer supervision—human teleoperation corrections or task-level objectives—which our recording infrastructure supports but this paper does not evaluate.

Noise robustness is unfinished. The temporal-difference channel should be trained under temporally correlated noise models fitted to the physical sensor, and the binary touch mask replaced by a soft (sigmoid) mask. Both are drop-in changes to the training pipeline.

Closed-loop evaluation. All results are open-loop replay: the policy’s commands do not feed back into the operator’s behavior. [PLACEHOLDER: Closed-loop user study: N participants guide the robot through reach/track tasks under (a) rule pipeline, (b) learned policy, on a healthy sensor and on a sensor with masked-off taxel regions; report completion time and path error. This is the strongest addition you can make before camera-ready.]

Scope. Rotational axes were locked to zero throughout; extending the corpus with two-finger rotation gestures is mechanical. Results come from a single sensor

Table 2. Effect of the location-preserving encoder and geometry-aware augmentation on noise robustness (velocity RMSE, m/s). The baseline variant uses a global-average-pooling encoder without geometric augmentation. Both variants are benchmarked identically. **[PLACEHOLDER: Re-run as a controlled single-variable ablation (same sequence length and training budget) before camera-ready; current numbers compare deployed variants that also differ in window length (8 vs. 16).]**

Variant	Clean	Noise $\sigma=0.25$	Noise $\sigma=0.5$	Dead 20%
GAP encoder, no geom. aug.	0.014	0.055	0.059	0.022
Spatial softmax + geom. aug. (ours)	0.014	0.033	0.043	0.019

unit and a small number of operators; **[PLACEHOLDER: multi-user, multi-unit generalization data if available].**

8 Conclusion

We showed that the hand-crafted rule pipelines conventionally used for continuous tactile robot guidance can be distilled into a compact learned policy—and, through a faithful-replay benchmark that exercises the deployed implementations of both controllers under identical calibrated corruptions, that the learned policy is the more robust of the two under baseline drift, dead taxels, and gain error, the failure modes that dominate capacitive skins in practice. The methodology is the more transferable half of the contribution: any tactile interface with a recorded teacher, rule-based or human, can be evaluated the same way. The learned policy’s noise weakness, traced to a temporal-difference input channel, illustrates precisely why such benchmarks should accompany every learned replacement of a working classical controller: on clean held-out data the two were indistinguishable, and only under perturbation did their differences—in both directions—become visible.

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